

1. Abstract

The Library Management System is a Java-based console application developed to simplify and organize common library operations. The system provides separate modules for managing books, students, book issue and return records, fines, and reports. It uses Object-Oriented Programming concepts to keep the application modular and easy to understand.

The application supports CRUD operations for books and students, checks book availability before issuing a book, maintains borrowing records, and automatically calculates fines for late returns. File handling is used to store records in text files so that data remains available between program sessions. The project demonstrates the practical application of Java concepts such as classes, objects, encapsulation, ArrayList, exception handling, file handling, and LocalDate.

2. Introduction

Managing library records manually can become difficult as the number of books, students, and transactions increases. A computerized system can make these activities more organized and reduce repetitive work.

This project provides a simple menu-driven library application that can be operated through the console. The user can manage book and student information, issue and return books, view borrowing history, handle fines, and generate basic reports.

3. Objectives

- To maintain book records in an organized manner.
- To maintain student registration and information.
- To provide a simple process for issuing and returning books.
- To automatically update available book copies.
- To maintain issue and borrowing history.
- To calculate and manage fines for late returns.
- To store records permanently using file handling.
- To demonstrate practical Java OOP and programming concepts.

4. Technologies and Concepts Used

Technology / Concept	Purpose
Java	Core programming language
Object-Oriented Programming	Classes, objects, encapsulation and modular design
ArrayList	Temporary handling of records
File Handling	Persistent storage of library data
Exception Handling	Handling invalid numeric input and file errors
LocalDate	Managing issue, due and return dates
VS Code	Development and execution environment

5. System Modules

- Book Management – add, view, search, update and delete book records.

- Student Management – add, view, search, update and delete student records.
- Issue / Return – issue books, return books and maintain issue records.
- Fine Management – view fines, view pending fines and record fine payments.
- Reports – display books, available books, issued books, students and fine records.
- File Management – create, read, write and append persistent text-file records.

6. Project Structure

The project is divided into model classes and manager classes. The model classes represent the main entities, while the manager classes handle the corresponding operations.

```
LibraryManagementSystem/
|
|— Main.java
|— Book.java
|— BookManager.java
|— Student.java
|— StudentManager.java
|— IssueRecord.java
|— IssueManager.java
|— Fine.java
|— FineManager.java
|— FileManager.java
|
|— data/
|   |— books.txt
|   |— students.txt
|   |— issue_records.txt
|   |— fines.txt
```

7. Object-Oriented Programming Implementation

The project applies OOP concepts by representing books, students, issue records and fines as separate classes. Data members are kept private and accessed through public methods, supporting encapsulation. Manager classes separate the application logic into modules, making the project easier to maintain and extend.

- Book and Student classes store entity-specific information.
- IssueRecord represents a book transaction.
- Fine represents a fine associated with an issue.
- Manager classes perform operations on the corresponding records.
- FileManager centralizes file creation and file read/write operations.

8. Implementation and Working

8.1 Main Menu

The application starts with a main menu that provides access to all major modules. The menu-driven design allows the user to navigate through the system using numbered choices.

```
=====
LIBRARY MANAGEMENT SYSTEM
=====

===== MAIN MENU =====
1. Book Management
2. Student Management
3. Issue / Return Book
4. Fine Management
5. Reports
6. Exit
=====
Enter your choice: █
```

Figure 1: Main menu of the Library Management System

8.2 Book Management – Add Book

The Add Book module collects the book ID, title, author, category and number of copies. After validation, the book is saved to the book data file.

```
===== BOOK MANAGEMENT =====
1. Add Book
2. View All Books
3. Search Book
4. Update Book
5. Delete Book
6. Back
Enter choice: 1

===== ADD BOOK =====
Enter Book ID: 101
Enter Book Title: The Housemaid
Enter Author: Freida McFadden
Enter Category: Mystery
Enter Number of Copies: 8
Book added successfully!
```

Figure 2: Adding a new book

8.3 Book Management – View Books

The View All Books option reads the stored book records and displays their details, including total and available copies.

```
===== ALL BOOKS =====  
-----  
Book ID      : 101  
Title       : The Housemaid  
Author      : Freida McFadden  
Category    : Mystery  
Total Copies : 8  
Available   : 8  
-----  
Book ID      : 102  
Title       : The Alchemist  
Author      : Paulo Coelho  
Category    : Fable  
Total Copies : 5  
Available   : 5  
-----
```

Figure 3: Displaying all book records

8.4 Student Management – Add Student

The system allows a student to be registered using a student ID, name, course, year and contact number. Duplicate student IDs are rejected.

```
===== ADD STUDENT =====  
Enter Student ID: 1058  
Enter Student Name: Paul Thomas  
Enter Course: CSE  
Enter Year: 2  
Enter Contact Number: 8897268769  
Student added successfully!
```

Figure 4: Adding a student record

8.5 Student Management – View Students

Registered student records can be displayed from the Student Management menu.

```
===== ALL STUDENTS =====  
-----  
Student ID : 1058  
Name       : Paul Thomas  
Course     : CSE  
Year       : 2  
Contact    : 8897268769  
-----  
-----  
Student ID : 1097  
Name       : Marvic  
Course     : ECE  
Year       : 3  
Contact    : 8723097867  
-----
```

Figure 5: Displaying student records

8.6 Issue Book

Before issuing a book, the system checks that both the book and student exist and that a copy is available. A successful transaction creates an issue record and updates the number of available copies. The application also assigns a due date.

```
===== MAIN MENU =====
1. Book Management
2. Student Management
3. Issue / Return Book
4. Fine Management
5. Reports
6. Exit
=====
Enter your choice: 3

===== ISSUE / RETURN =====
1. Issue Book
2. Return Book
3. View Issue Records
4. Student Borrowing History
5. Back
Enter choice: 1

===== ISSUE BOOK =====
Enter Book ID: 102
Enter Student ID: 1097

Book issued successfully.
Issue ID : 1
Due Date : 2026-10-01
```

Figure 6: Successfully issuing a book

8.7 Return Book

The Return Book option uses the issue ID to locate the transaction. After a successful return, the book's available-copy count is updated and the issue record is marked as returned.

```
===== ISSUE / RETURN =====
1. Issue Book
2. Return Book
3. View Issue Records
4. Student Borrowing History
5. Back
Enter choice: 2

===== RETURN BOOK =====
Enter Issue ID: 1
Book returned successfully.
```

Figure 7: Successfully returning a book

8.8 Reports and Issue Records

The Reports menu provides access to all books, available books, issued books, students and fine records. The issue records section shows the book, student, issue date, due date, return date and current status of transactions.

===== REPORTS =====

1. All Books
2. Available Books
3. Issued Books
4. Students
5. Fine Report
6. Back

Enter choice: 1

===== ALL BOOKS =====

Book ID : 101
Title : The Housemaid
Author : Freida McFadden
Category : Mystery
Total Copies : 8
Available : 7

Book ID : 102
Title : The Alchemist
Author : Paulo Coelho
Category : Fable
Total Copies : 5
Available : 5

===== REPORTS =====

1. All Books
2. Available Books
3. Issued Books
4. Students
5. Fine Report
6. Back

Enter choice: 3

===== ISSUE RECORDS =====

Issue ID : 1
Book ID : 102
Student ID : 1097
Issue Date : 2026-09-17
Due Date : 2026-10-01
Return Date : 2026-09-17
Status : RETURNED

9. File Handling and Data Storage

The application uses text files for persistent storage. At startup, the FileManager creates a data directory and the required files if they do not already exist. Separate files are used for books, students, issue records and fines.

- books.txt – stores book information.
- students.txt – stores student information.
- issue_records.txt – stores issue and return transactions.
- fines.txt – stores fine records and payment status.

This approach allows records entered during one session to remain available when the application is opened again.

10. Input Validation and Error Handling

The project includes input validation for numeric fields and text fields. Positive integer values are checked before being accepted, while invalid numeric input is handled using NumberFormatException. Text input is also checked so that empty values are not stored.

- Prevents duplicate book and student IDs.
- Rejects invalid numeric input.
- Prevents empty text fields.
- Prevents the '|' separator from appearing inside stored text values.
- Prevents deleting a book while copies are currently issued.
- Prevents issuing a book when no copies are available.

11. Testing

Test Case	Expected Result	Observed Result
Add Book	Book record is saved	Successful
View Books	Saved books are displayed	Successful
Add Student	Student record is saved	Successful
Issue Book	Issue record created and availability updated	Successful
Return Book	Return date/status updated and availability restored	Successful
View Reports	Requested records are displayed	Successful

12. Results

The completed application successfully demonstrates the main operations required for a basic library management system. The screenshots included in this report show the working console interface, book and student management, book issuing and returning, and report generation.

13. Conclusion

The Library Management System provides a simple and functional way to manage common library activities through a Java console application. The project helped demonstrate the practical use of Object-Oriented Programming, file handling, collections, exception handling, input validation and date handling. The modular structure also makes the application easier to understand and provides a base for future improvements such as a graphical interface or database integration.

14. Future Scope

- Develop a graphical user interface using JavaFX or Swing.
- Replace text-file storage with a database such as MySQL.
- Add user authentication and separate admin/student access.
- Add online book reservation and availability notifications.
- Add barcode or QR-code support for books.
- Generate more detailed reports and dashboards.

15. References

- Java programming concepts and standard library documentation used during development.
- VS Code as the development environment.
- Project source code developed for the Library Management System.